



MBA (Trimester-II) Regular Examination February 2026

Name of the Course MBA End Semester Examination 2024

Semester: Trimester II

Subject: Name Supply Chain Management

Paper Code: MBA202

Time: 3:00 Hours

MM: 100

Note:

- (i) This question paper contains two Sections-Section A and B
- (ii) Both Sections are compulsory
- (iii) Answer any two sub questions from a, b & c in each main question of Section A. Each sub question carries 10 marks.
- (iv) Section B, consisting of a case study, is compulsory. It is of 40 Marks.

Section A

- Q1 (10 X 2 Marks) COs
- a) Explain the concept of Supply Chain Management and discuss its scope in the context of modern competitive markets. How does SCM contribute to organizational value creation? CO1
 - b) Compare push-based and pull-based supply chain strategies. Evaluate their suitability across different product types and market conditions. CO2
 - c) Evaluate alternative sourcing strategies such as multi-sourcing, near-shoring, and strategic partnerships. Which sourcing mix would you recommend and why? CO2, CO5
- Q2 (10 X 2 Marks)
- a) A manufacturing company has an annual demand of 15,000 units for a particular component. The ordering cost is ₹ 750 per order, and the annual holding (carrying) cost is ₹ 25 per unit. Calculate the CO3
 - (i) Economic Order Quantity (EOQ).
 - (ii) Determine the number of orders placed per year and the time interval between orders
 - (iii) Interpret the managerial significance of EOQ in inventory decision-making. How does EOQ help balance ordering and holding costs?
 - b) Which supply chain drivers should a firm prioritize in order to reduce its cash conversion cycle? Between a traditional brick-and-mortar retailer and an online retailer, which business model would you expect to exhibit a higher asset turnover, and why? CO2
 - c) Analyze the trade-offs between centralization and decentralization in supply chain network design. Under what market conditions is each approach strategically superior? CO3



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Q3

(10 X 2 Marks)

- a) A company selling air coolers recorded the following monthly demand (in units): CO3, CO4

Month Demand

Jan	420
Feb	450
Mar	470
Apr	500
May	530
Jun	560

- a) Calculate the 3-month moving average forecast for April, May, June, and July.
- b) Using simple exponential smoothing with $\alpha = 0.3$ and assuming the forecast for January equals actual January demand, calculate the forecast for July.
- c) Compare the forecasting results from both methods and comment on which method is more responsive to demand changes.
- b) Critically assess the role of inventory as a strategic buffer versus a source of inefficiency. Is zero-inventory a realistic or desirable goal? CO3
- c) Analyze the relationship between sustainability and profitability in supply chains. Are green supply chain practices a cost burden or a source of long-term competitive advantage? CO5

Section – B

Q4 Case Study

(40 Marks)

CO5, CO6

FreshMart Retail Ltd. is a mid-sized supermarket chain operating across 25 cities in India. The company deals in fast-moving consumer goods (FMCG), packaged food items, household products, and limited fresh produce. Over the past year, FreshMart has experienced rising inventory carrying costs, frequent stockouts of high-demand products, and increasing product obsolescence in slow-moving categories.

Currently, FreshMart follows a traditional replenishment system where store managers manually place orders based on past experience and visual stock checks. The company maintains relatively high safety stock levels to avoid stockouts, particularly during festival seasons. However, this has led to excessive working capital blockage and higher warehousing expenses.

An internal audit revealed the following issues:

- Demand forecasting is inconsistent and not data-driven.
- Lead times from suppliers vary significantly.
- Some SKUs have high holding costs but low turnover rates.



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- There is limited coordination between procurement, warehousing, and store operations.

Senior management is considering implementing centralized demand forecasting, automated replenishment systems, and an EOQ-based ordering approach. They are also evaluating whether certain slow-moving SKUs should be discontinued. The management has appointed you as a supply chain consultant to diagnose the problem and recommend solutions.

Q1. Analyze the key inventory management problems faced by FreshMart. How are excessive inventory and stockouts occurring simultaneously? Explain the operational and financial implications of these issues.

Q2. Evaluate the suitability of inventory management tools such as EOQ, ABC analysis, and safety stock optimization for FreshMart. How can these tools improve cost efficiency and service levels?

Q3. Discuss how centralized demand forecasting and automated replenishment systems could improve inventory performance. What challenges might FreshMart face during implementation?

Q4. Develop an integrated inventory management strategy for FreshMart that balances cost control, service level improvement, and working capital efficiency. Justify your recommendations.